

ICT Skills Studio

Computing and Mathematics

May, 2026

Short Title:	ICT Skills Studio	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Software and Web Development	
Author	EDELEASTAR	
Credits	5	Level: Intermediate

Description of Module / Aims

Continue to develop the students' programming skills with a particular focus on modern web development tools, frameworks and applications. A strong emphasis industry best practice - with close attention software configuration management tools. The Studio will also support a forum for talks and seminars from industry on technology trends and work practices.

Programmes

COMP-0521 Higher Diploma in Science in Computer Science (WD_KCOSC_G)	1 / 2 / M
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Indicative Content

- Consolidating programming principles & practices
- Elementary algorithms & data structures
- Web development applications design & implementation
- Server side and client side aspects of web development
- Current trends in software development technology

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Construct a simple but robust server-rendered web application.
2. Differentiate the role of the client and server in this context.
3. Break a problem domain into a series of discrete features.
4. Manage the assets of a project in a configuration management environment.

Learning and Teaching Methods

- Supervised, guided and scripted practicals will lead the student through the construction of selection of small applications, designed to illustrate key concepts covered in the lectures.
- Worked Laboratory Exercises.
- Short Lectures to outline specific concepts.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Assignment</i>	100%	1,2,3,4

Assessment Criteria

<40%: Unable to interpret, describe or implement key components of a simple web application.

40%-49%: Be able to construct a minimal web application, incorporating server side rendering.
50%-59%: Design and implement a web application including basic database and session support.
60%-69%: All of the above including a simple model of the application structure.
70%-100%: All the above to an excellent level. Incorporate Unit Tests for a specific subset of an application.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Supplementary Material

- Richard-Foy, J. *_Play Framework Essentials_*. New York: Packt Publishing, 2014.
- Syed, B. *_Beginning Node.js_*. New York: Apress, 2014.

Requested Resources

- COMPUTER LAB: BYOD Lab